THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM TWO PRE – MOCK EXAMINATON
CHEMISTRY

TIME: 2:30 Hrs

Instructions.

- Attempt all questions in section A,B and C
- All writtings must be in blue or black ink.
- The following constants may be used.

Atomic masses.

H =1 , N= 14 Ca =40 , O =16 , Zn =65, Cl = 35.5, S = 32, Na = 23, C= 12, Mg = 24

SECTION A (10 MARKS)

1. For each of the following items (i – x) choose the correct answer from the given alternatives and write its letter in the box provided.

i. Group seven elements are known as _____

A. Alkaline metal B. Transitional elements C. Alkaline Earth metal D. Halogens

ii. A Bunsen burner is likely to contain soot when

A. The air holes is closed B. The hunsen burner is closed C. The air hole is open

D. The gas supply is higher

iii. The arrangement of electron in the atom of atomic number 15^0 may represented as follows

A. 2:5:8 B. 2:8:5 C. 8:5:2 D. 8:2:5

iv. A businessman found that a significant amount of sugar is mixed with sand without its knowledge. Which appropriate method will he use in a sequence to recover the sugar?

A. Dissolution, decantation, filtration and evaporation.

B. Decantation, filtration and evaporation.

C. Dissolution, decantation, filtration and solvent extraction.

D. Clamatography, filtration and evaporation

v. Which element will form a compound of the formula M_2O_3, where M is a metal?

A. Oxygen and beryllium B. Oxygen and sodium

C. Oxygen and sulphur D. Oxygen and aluminium

vi. Amina left an Ordourless and colourless liquid on her work banch. A limus paper wa dipped into the liquid but gave no change, alittle of the liquid was dropped into anhydrous copper (II) sulphate and it turned $Cuso_4$ into blue. The liquid is likely to be:,

A. Ethanol B. Methanol C. water D. sodium hydroxide solution

vii. The total number of protons and neutrons in the nucleus of an atom is called?

A. Vallency number B. Atomic number C. Molecule number D. Mass number

viii. When an atom losses or gain electron, it becomes _____

A. An iron B. anion C. neutral charged ion D. Cation

ix. Which of the following is not among the gasses composing air?

A. Noble gases B. Carbon dioxide C. Nitrogen D. Hydrogen

x. Which of the following set of nuclide notation represents isotopes?

A.

2. Match each item in List A with a correct response in List B by writing its letter below the number of the corresponding item in the table provided.

LIST A	LIST B
i. Group of elements which reacts quickly with water to form alkaline solution	A. Metalloids
ii. The ability of an atom to attract an electron	B. Non – metals
iii. Groups of elements in which their shells are completely filled up.	C. Periodicity
iv. A vertical column of elements in the	D. Transition elements
	E. Electron negativity
	F. Alkali metals
	G. Halogens

	periodic table.	H. Periodic law
v.	Groups of elements which reacts with metal to form salts	I. Alkali Earth metals
		J. Rare non metals
vi.	Groups of elements which reacts slowly with water to form alkaline solution	K. Period
		L. Noble gases
vii.	Group of elements which reacts by receiving electrons	M. Periodic table
		N. Group
viii.	The systematic arrangement of elements according to their increase in atomic number.	
ix.	Group of elements with high densities and melting paints and often act as catalysts.	
x.	Group of elements which have both metallic and non – metallic characteristics	

SECTION C. (70 Marks)

3. (a) Calculate the oxidation number of the underlined elements.

(i) $K_2C_2O_4$	(ii) SO_3^{-2}

(b) Write the use of the following apparatus;

i. Thistle funnel _____
ii. Pipettle _____
iii. Wire Gauze _____
iv. Burettle _____

4. (a) Define the following terms;

i. Oxidation state _____
ii. An element _____
iii. A compound _____
iv. Fainting _____

(b) State two chemical properties of water

i. _____
ii. _____

(c) Give the differences between;

Solutions	Suspensions

5. Gas X can be prepared in the laboratory by the decomposition of hydrogen peroxide.

(a). Identify gas "X" _____

(b) State two physical properties of gas "X"

i. _____
ii. _____

(c) Mention three chemical properties of gas " X"

i. _____
ii. _____
iii. _____

(d) State three uses og gas " X"

i. _____
ii. _____
iii. _____

6. (a) Study the following periodic table and then answer the questions that follow;

I							VIII
A	II	III	IV	V	VI	VII	B
	C		D		E		
F	G					H	

(i) Write the collective name of elements in ;

Group II _____

Group VIII _____

(ii) Name the elements which are represented by the following letters ;

A. _____ B. _____ C. _____ D. _____

(iii) Give the electronic configuration of the following elements;

E. _____ G._____

F. _____ H. _____

(b) Name two products in each of the following fields made by the applicationnof chemistry.

Field	Products

(a) Medicine	
(b) Food and Bererage industry	

7. When dilute hydrochloric acid is reacted with zin metal, Gas Z is formed.

(a) Identify the name of gas Z

_____

(b) Mention four (4) physical properties of gas Z

i. _____
ii. _____
iii. _____
iv. _____

(c) State two uses of gas Z

i. _____
ii. _____

8. (a) Draw an electronic configuration to illustrate the structure of atoms P and Q with 13 and 17 protons.

(b) From the structure drown in (a) above, state the vallency, period and group the atoms are placed ina periodic table.

P- Valency is _____

Period is _____

Group is _____

Q - Vallency is _____

Period is _____

Group is _____

9. (a) Write the symbols for the following elements;

i. Silver _____
ii. Copper _____
iii. Iron _____
iv. Mercury _____

(b) What is the meaning of covalent bonding?

_____
_____

(c) Mention four differences between covalent and electrovalent compounds.

Covalent compounds	Electrovalent compunds

10. (a) What is a flame? _____

(b) Name two types of flames

i. _____
ii. _____

(c) A burning flame was observed producing soot and much light.

i. Name the type of the flame _____

_____

ii. State the colour of the flame _____

iii. How many zones does the flame in (c) (i) has? _____

iv. By using the Bunsen burner, state how you can produce such a flame type

_____

_____

v. Draw a diagram of the flame in (c) (i)